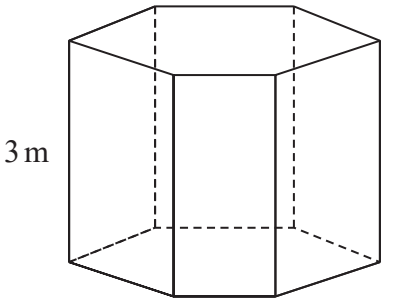
27 The diagram shows a prism placed on a horizontal floor.



$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The prism has height 3 m
The volume of the prism is 18 m^3

The pressure on the floor due to the prism is 75 newtons/m^2

Work out the force exerted by the prism on the floor.

Volume = base area × height.

Area touching floor = 18 ÷ 3 = 6 m^2.

Pressure = force ÷ area, so force = 75 × 6 = 450 N.

3 marks total
Base area 6 m^2
Use pressure × area
450 N

..... newtons

(Total for Question 27 is 3 marks)

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